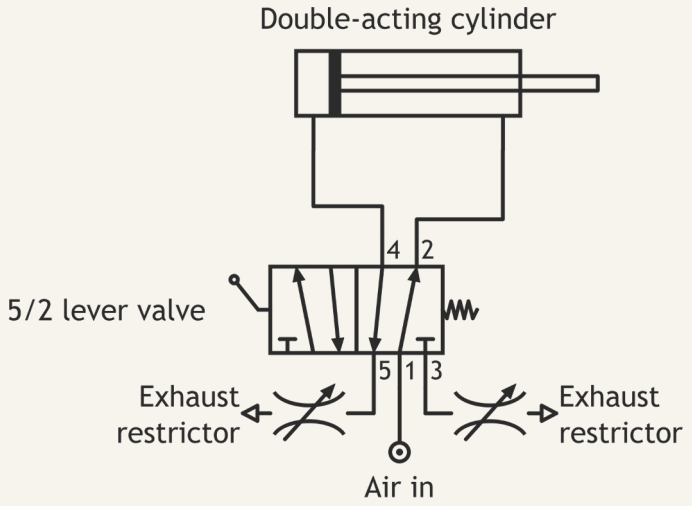
Daily Review

1	What does a flow control valve slow?	The air, one way only
2	Arrow to the cylinder: the rod goes out…	Slowly
3	What moves a 3/2 button back?	Its spring
4	A lever on a system diagram is an…	Input
5	A push or a pull is called a…	Force

Push and Pull

What if air pushes the rod out AND back in?



S3 Engineering Science

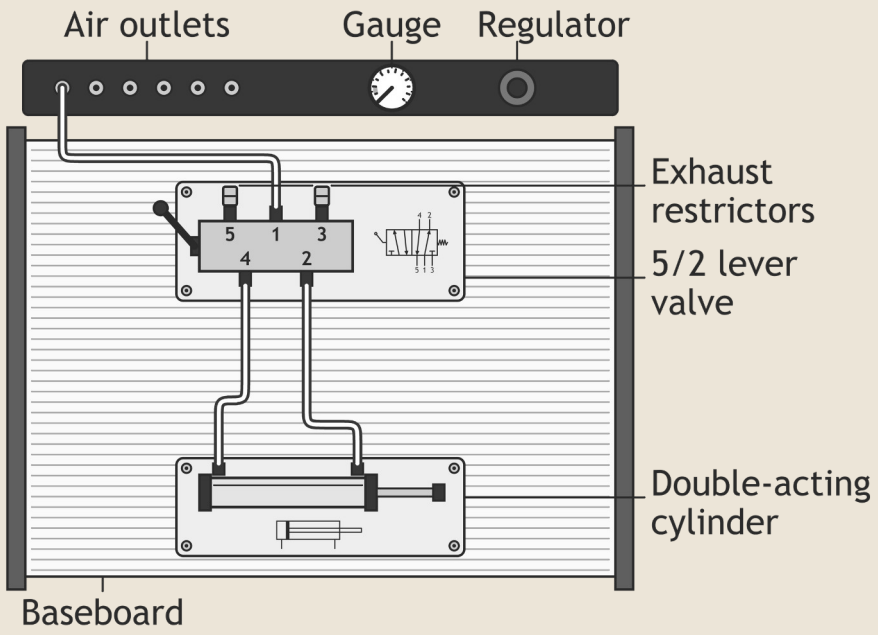
The Job: The Paint Dip

i When the lever is pushed, the basket dips.

ii When the lever is let go, air lifts the basket.

iii When the restrictors are set, it dips and lifts slowly.

Predict: which restrictor slows the dip, 3 or 5?



By the end of this lesson…



1. I can tell a single-acting from a double-acting cylinder
2. I can build a circuit with a 5/2 valve
3. I can set a rod's speed with exhaust restrictors

THE BUILD

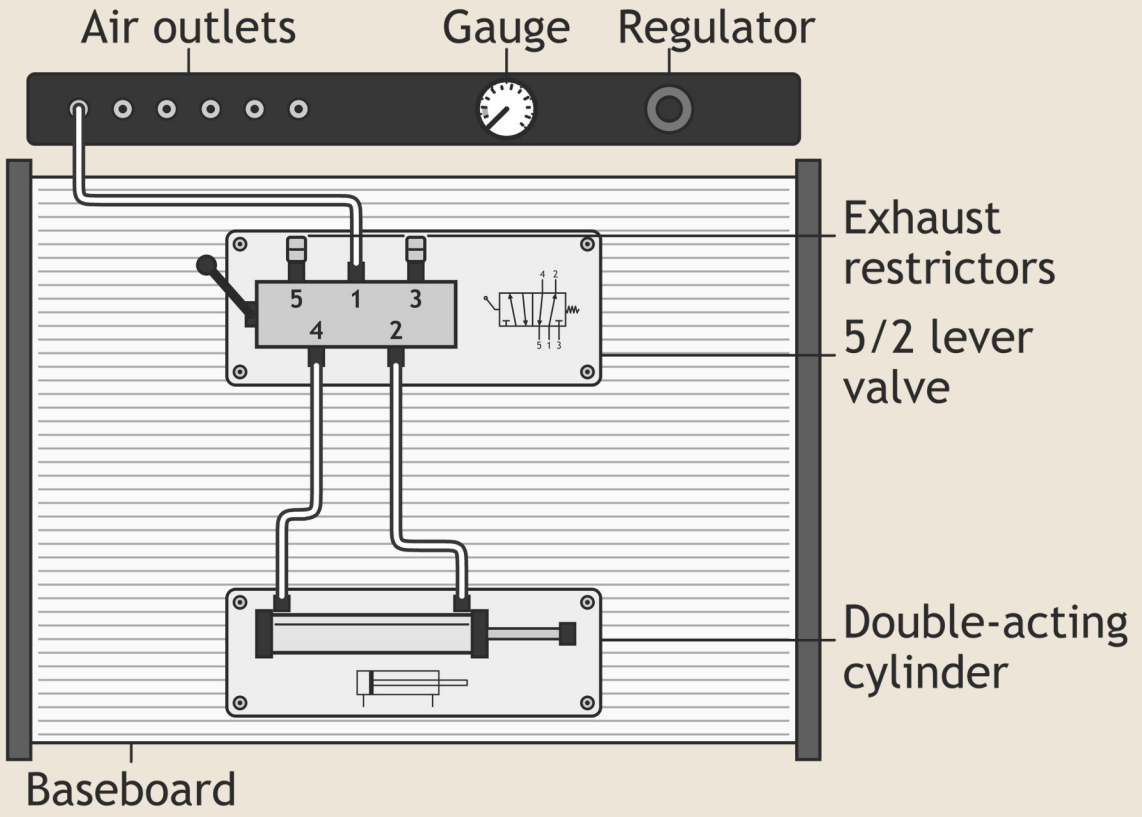
Build the Paint Dip

Feel it. Simulate it. Air Permit. Then the kit.

What to Collect

Builder and Safety: one set per team.

1 Bench card 4
2 Manifold (air supply)
3 5/2 lever valve
4 Double-acting cylinder
5 Single-acting cylinder
6 2 exhaust restrictors
7 Tubing
8 Goggles, one each

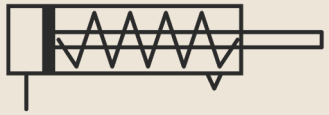


Feel the Difference

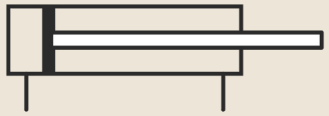
Task 1 (B6A-L4-p11)

Pull each rod out by hand, then let go.
Write what happens in the table.

Single-acting: one port, a spring pushes it back



Double-acting: two ports, air pushes both ways

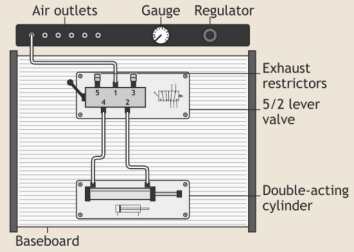


SAFETY Air off. Regulator at 0. No tubes in.

3 min

Simulate and Build

Tasks 2-3 (B6A-L4-p12)

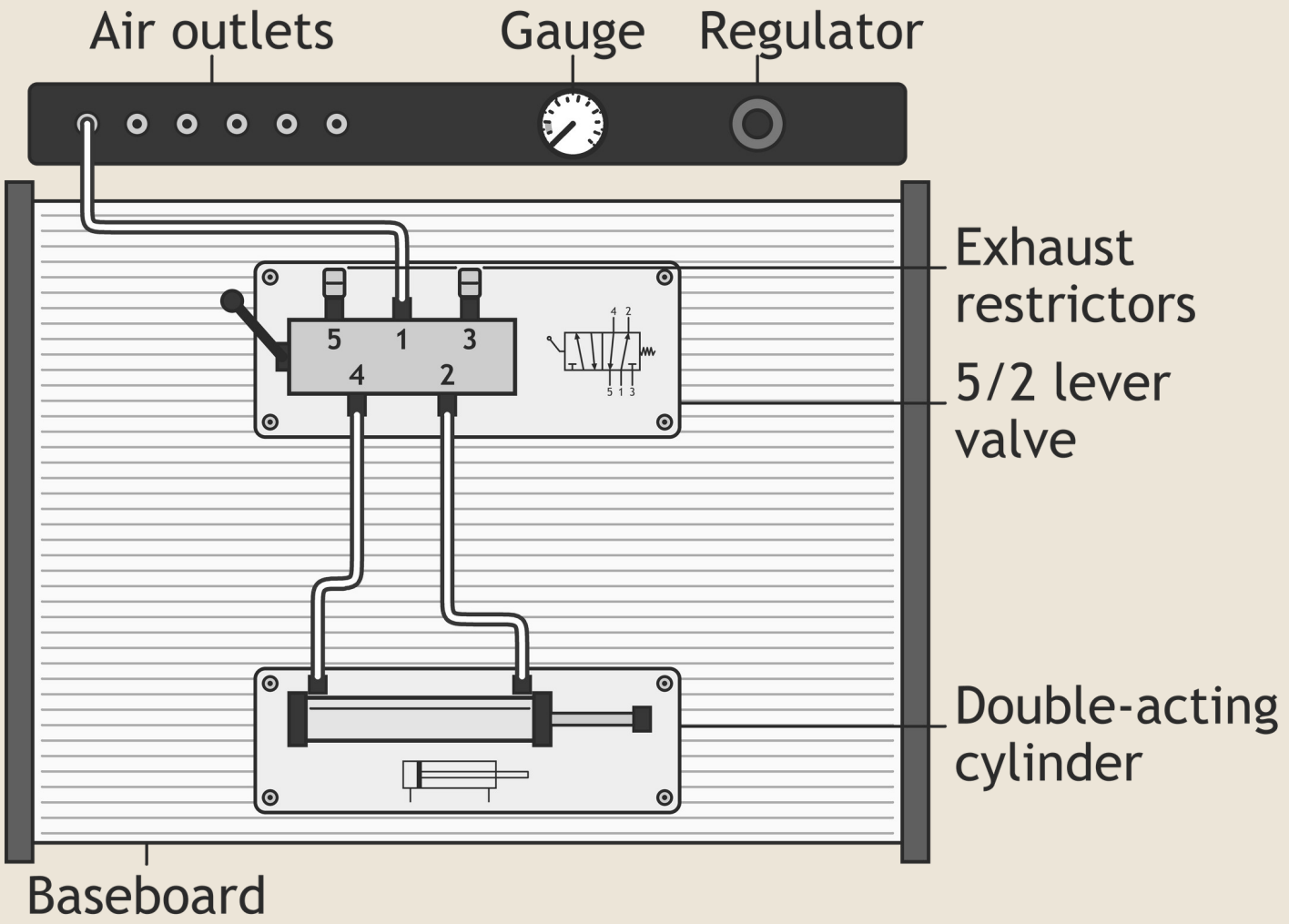


Tester and Recorder: start on the computer.
Builder and Safety: parts and bench card 4.
Safety, then teacher: check the Air Permit.

SAFETY Regulator at 0 to build. 1 to 2 bar to test.

14 min

Build It: Bench Card 4



Find Out: Set the Speeds

Task 4 (B6A-L4-p13)



Turn port 3's restrictor in a little. Test.
Tick the way the rod slowed.
Then do the same with port 5.

SAFETY Turn a restrictor only with the rod still.

5 min

Test It: What Did You Find?

1	Let go: the single-acting rod… the double?	Springs back; stays out
2	Points i and ii met? What pushed it back?	Yes: air, not a spring
3	Port 3 turned in. Which stroke slowed?	Out: the dip
4	Port 5 turned in. Point iii met?	The lift slowed: yes

One Port or Two

Single-acting

one port and a spring

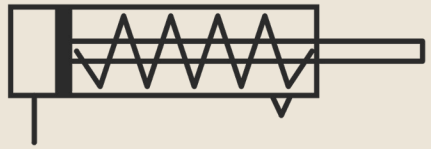
Double-acting

two ports: air both ways

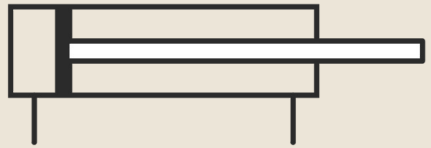
Outstroke, instroke

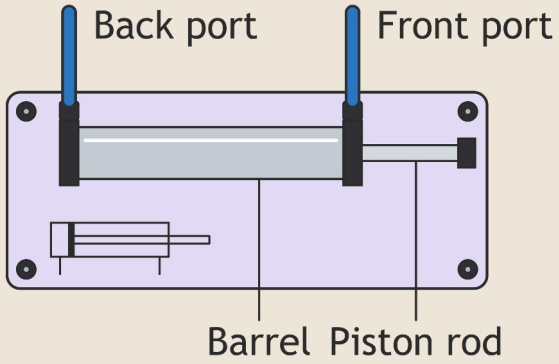
the rod moves out, then in

Single-acting: one port, a spring pushes it back



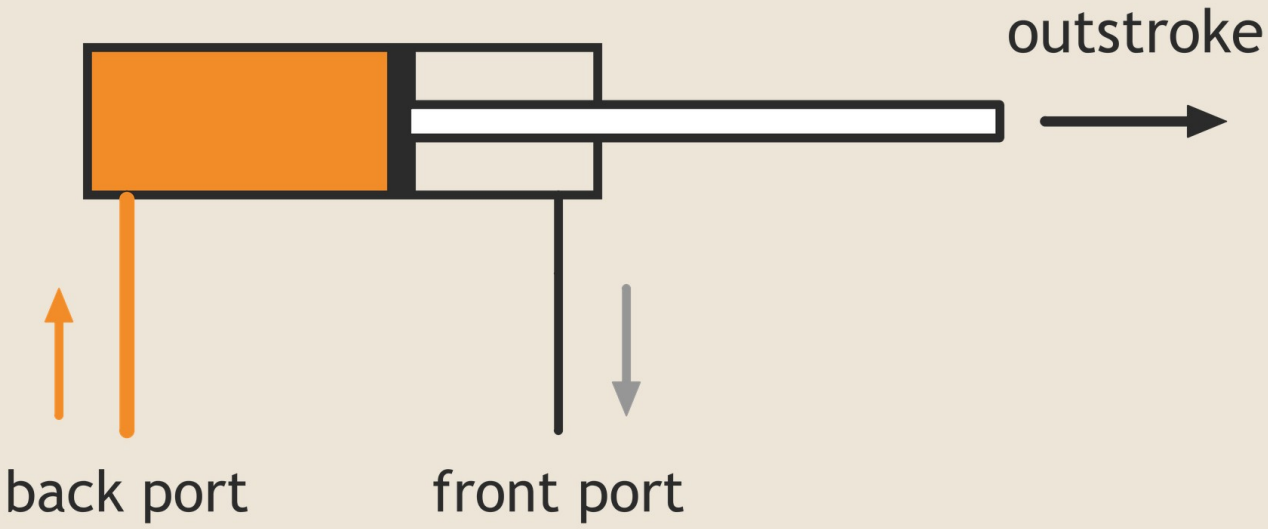
Double-acting: two ports, air pushes both ways





The real double-acting cylinder.

Worked Example: The Dip

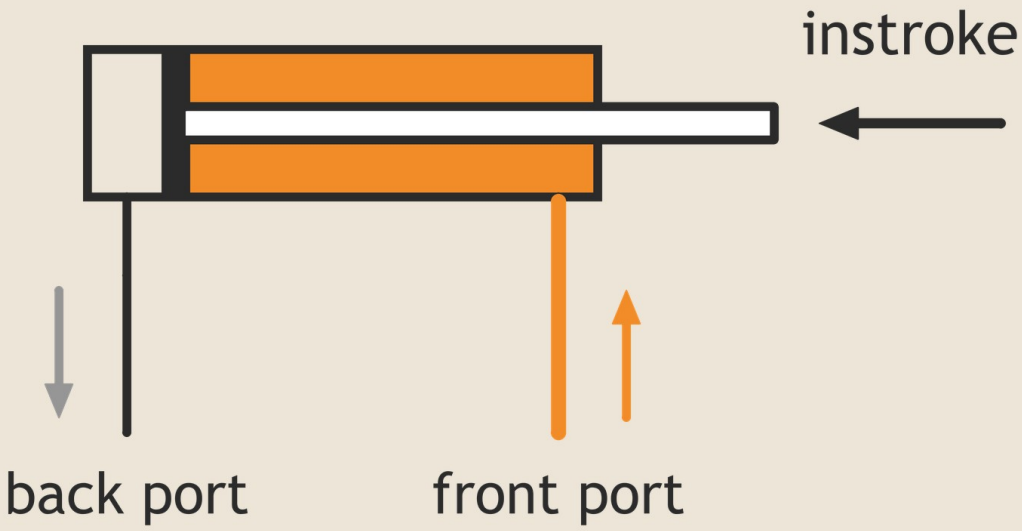


1 Air goes in at the back port.

2 The back fills; the front air leaves.

3 The rod moves out: outstroke.

We Do: The Lift



1 Air goes in at the front port.

2 The front fills; the back air leaves.

3 The rod moves in: instroke.

One More Push: the Plunger

Plunger

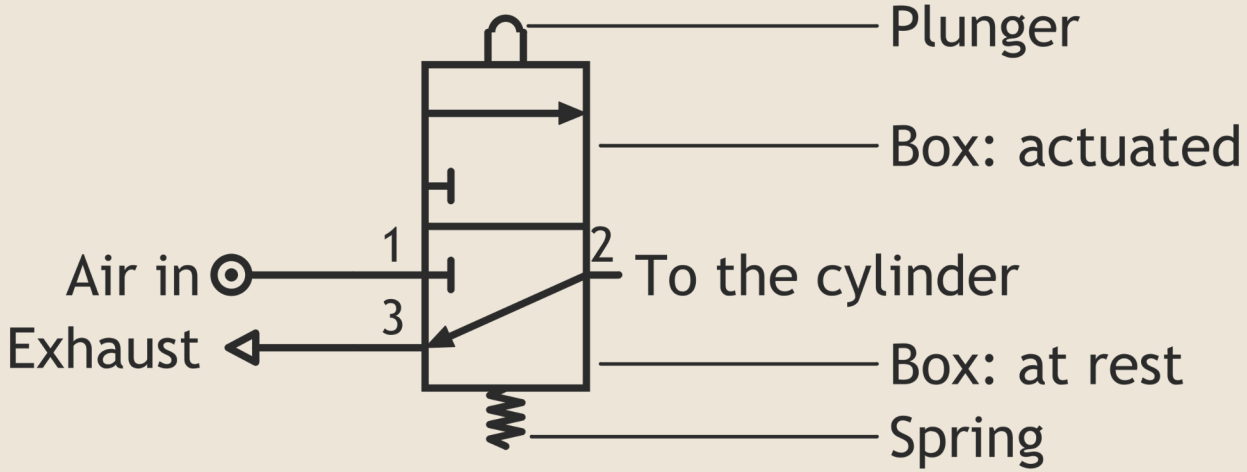
a short rod on top

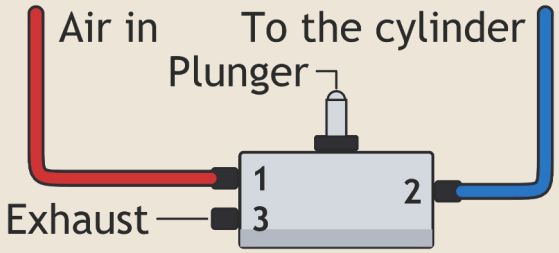
Pressed by

a moving part

Symbol

a rounded rod on top





The real plunger valve.

The 5/2 Valve

Level Up

5/2

5 ports, 2 states

1

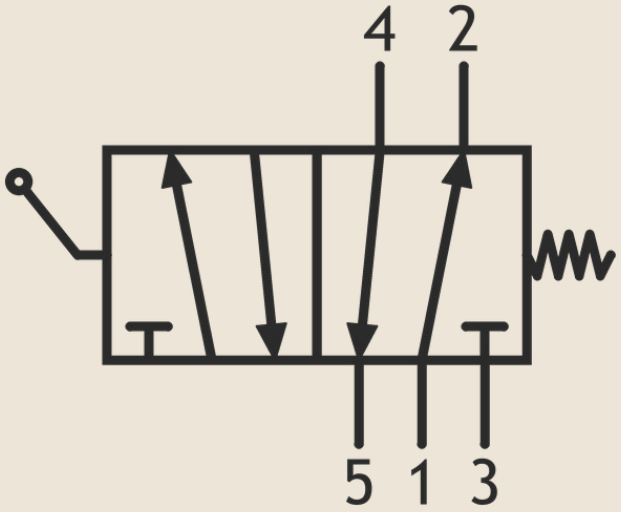
air in

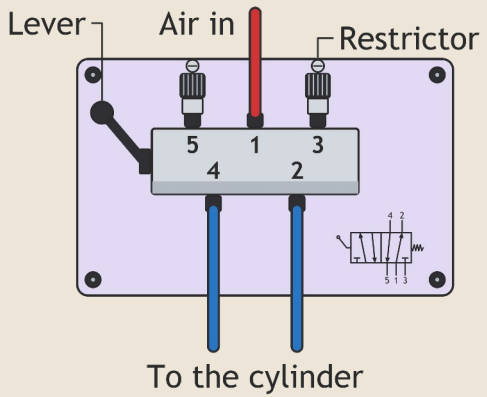
2 and 4

out to the cylinder

3 and 5

exhausts

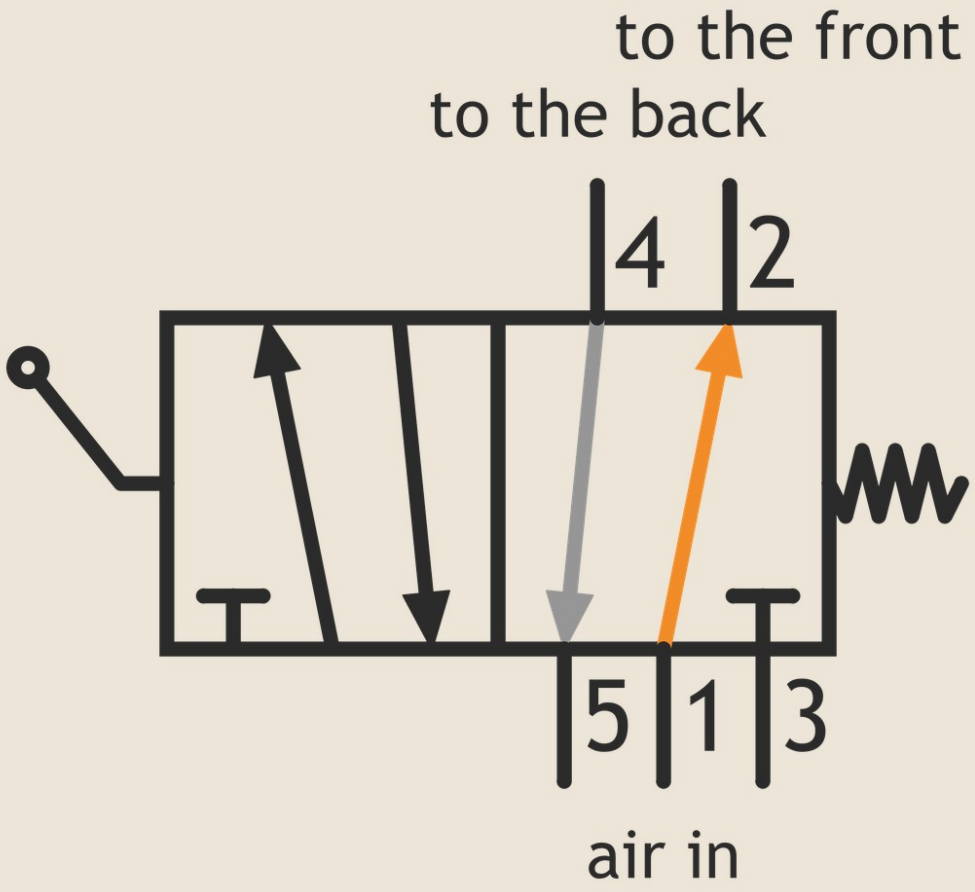




The real 5/2 lever valve.

Push the Lever, Then Let Go

Level Up



1 Push: the boxes slide. Air 1 to 4: the back.

2 Let go: the spring. Air 1 to 2: the front.

The Circuit You Built

Level Up

Input

5/2 lever valve

Port 4

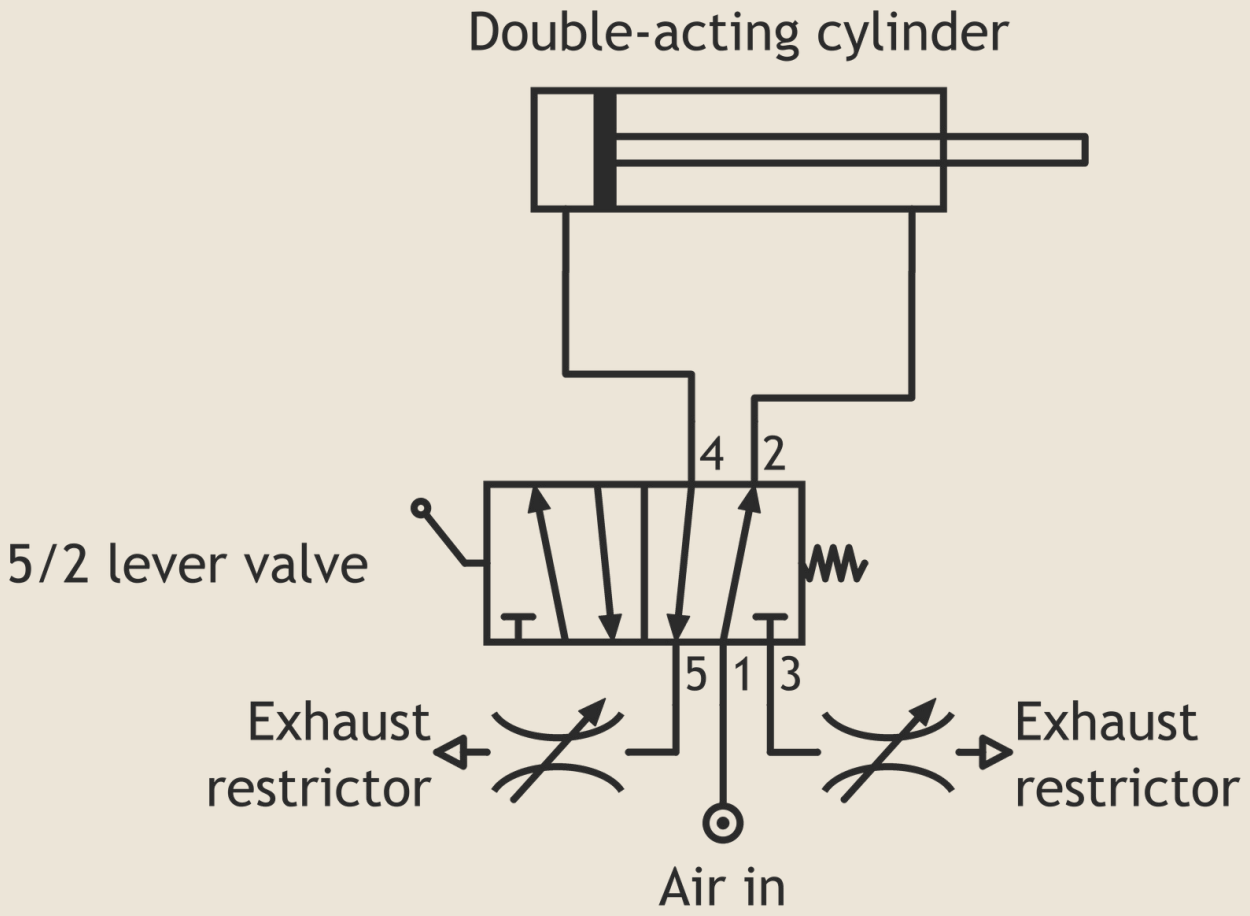
to the back

Port 2

to the front

Output

the cylinder



Exhaust Restrictors

Level Up

Fits

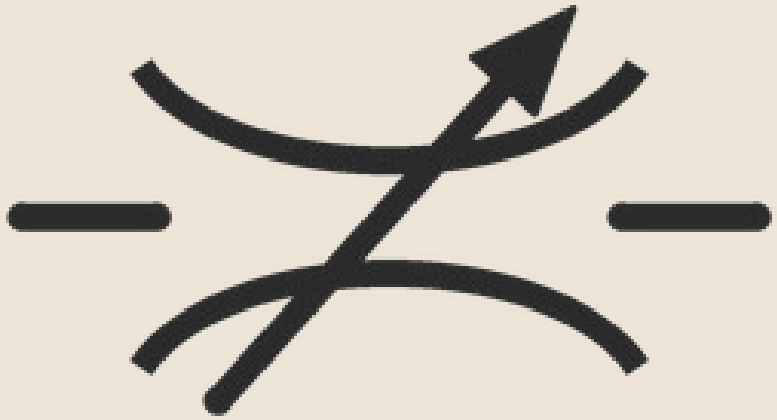
exhaust port 3 or 5

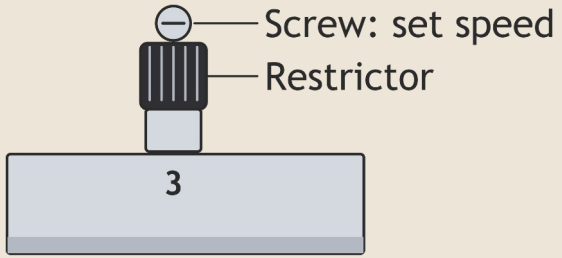
Turn it in

air leaves slowly

So

the rod slows





Fits in an exhaust port.

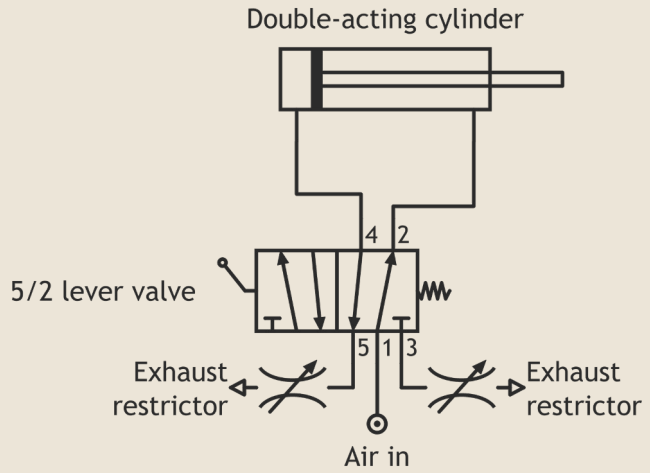
The real exhaust restrictor.

Compare and Questions

Tasks 5-6 (B6A-L4-p13)

Task 5: fill in Compare.

Task 6: Questions 1 to 3 on your own.



5 min

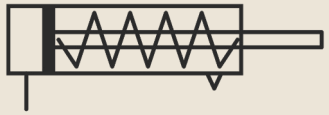
Key Words and Exam Practice

Tasks 7-8 (B6A-L4-p14)

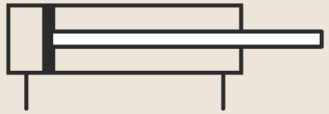
Task 7: Key words.

Task 8: the logo stamp question.

Single-acting: one port, a spring pushes it back



Double-acting: two ports, air pushes both ways



4 min

Answers: Tasks 6-7 (B6A-L4-p13-14)

Question	Mark 1	Mark 2
Q1 (1 mark)	One port or two	-
Q2 (2 marks)	Heavy basket	Spring cannot lift it
Q3 (1 mark)	Slows air one way	-
Key words 1-2	one; spring	two; Air
Key word 3	outstroke	-

Answers: Task 8 (B6A-L4-p14)

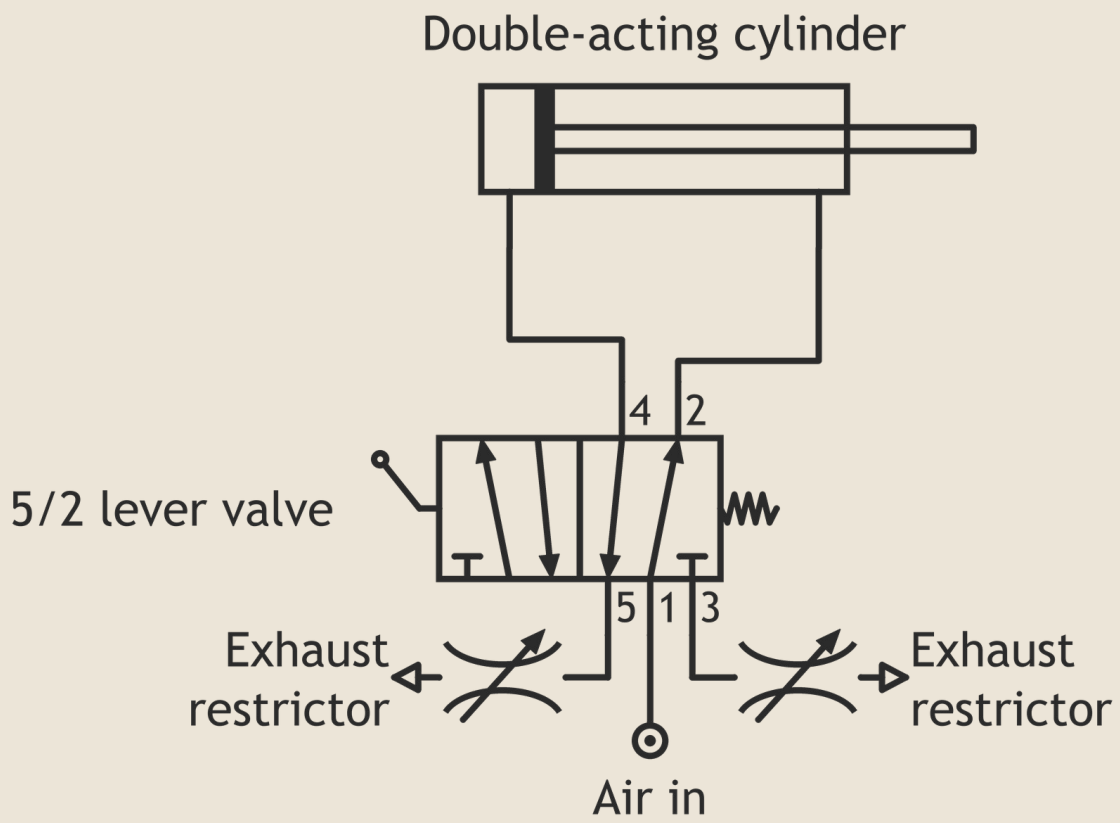
Level Up

Restrict the air (1)

fit an exhaust restrictor

Where (1)

on the exhaust the front air leaves by: port 3



Plenary: Push and Pull

1	Ports on a double-acting cylinder?	2
2	The rod moves in. The exam word?	Instroke
3	Double-acting: what pushes it back?	Air
4	Why double-acting for the dip?	A spring cannot lift it
5	Level Up: which exhaust slows the dip?	Port 3

Look back — by the end of this lesson…



1. I can tell a single-acting from a double-acting cylinder
2. I can build a circuit with a 5/2 valve
3. I can set a rod's speed with exhaust restrictors

LOOKING AHEAD

Lesson 5: Will It Lift?

How heavy a basket can the air lift?

Credits

The job and the kit: Sources: job card and kit drawings by R Stewart.

Paint dip circuit: The Lesson 4 circuit adapted from the Matrix CP2080 Pneumatics Essentials Worksheet 4 (© Matrix Technology Solutions Limited). Drawings by R Stewart.

Real part drawings: Sources: kit drawings by R Stewart, from the Matrix CP2080 kit.

Lesson 4 circuit and kit pictures: The Lesson 4 circuit adapted from the Matrix CP2080 Pneumatics Essentials Worksheet 4 (© Matrix Technology Solutions Limited). Drawings by R Stewart.

Task 8 and Model Answer: 2024 Q15(c): © Qualifications Scotland (SQA). N5 Engineering Science 2024, Question 15(c) and its marking instructions (booklet Task 8, adapted).

R Stewart - October 2026